Expectation Decider

Probability Analysis Report - Poori Explanation (Hinglish)

Mathematics & Advanced Statistics | Project PR1

Name	YOUR NAME
GRID	YOUR GRID
Dataset	student_data.csv (200 students, AI-generated)
Fields	study_hours, attendance, group_discussion, previous_test_score, final_exam_pass
Tools	Python, Pandas, Matplotlib, SciPy (Jupyter Notebook)

Project ka maksad: 200 students ke data se yeh samajhna ki kaun se factors (study hours, attendance, group discussion, previous test score) ek competitive maths exam mein pass hone ki **probability** ko sabse zyada affect karte hain.

Is PDF mein kya hai: Har task (1 se 7) ke liye pehle concept simple bhasha mein, phir formula, phir calculation aur aakhir mein result ka matlab. Sab numbers aapke *student_data.csv* aur notebook ke output se liye gaye hain.

Dataset ka overview

Field	Matlab	Type
study_hours	Ek hafte mein kitne ghante padhai ki	Number
attendance	Lectures mein attendance (%)	Number
group_discussion	Group discussion mein hissa liya? (Yes/No)	Category
previous_test_score	Pichle internal test ke marks (out of 100)	Number
final_exam_pass	Competitive exam mein Pass ya Fail	Category

Total **200 students**, koi missing value nahi. In mein se **124 pass** hue aur **76 fail**.

Task 1: Probability ki Basics

Probability kya hai?

Probability ek number hai (0 se 1 ke beech) jo batata hai ki koi event hone ke kitne chances hain. 0 ka matlab **impossible**, 1 ka matlab **pakka hoga**. Jitna 1 ke paas, utne zyada chances.

Formula: P(A) = (Favourable outcomes) / (Total outcomes)

Important terms

Term	Simple matlab	Hamare dataset se example
Experiment	Aisi activity jiska result pehle se pata na ho	Ek student random chunna
Outcome	Ek possible result	Student #57 chuna gaya
Sample Space (S)	Saare possible outcomes	Saare 200 students
Event	Sample space ka ek hissa	Wo students jo pass hue (124)
Complement	Event hua hi nahi: P(A') = 1 - P(A)	Fail hone ki probability = 1 - 0.62 = 0.38
Mutually exclusive	Dono saath mein nahi ho sakte	Pass aur Fail (ek student ek hi ho sakta hai)
Independent events	Ek ka hona doosre ki probability nahi badalta	Do alag logon ka coin toss

Dataset se 3 probability event examples

#	Event	Calculation	Probability
1	Random student pass hua	124 / 200	0.620 (62%)
2	Random student 10 ghante se zyada padhta hai	119 / 200	0.595 (59.5%)
3	Random student ka previous score 75 ya usse zyada	34 / 200	0.170 (17%)

Task 2: Empirical vs Theoretical Probability

- **Empirical probability:** Jo data mein actually dekha, uske basis par. Formula: (event kitni baar hua) / (total trials).
- **Theoretical probability:** Jab saare outcomes barabar likely hon, tab bina data dekhe sirf logic se nikalte hain.

Type	Scenario	Calculation	Result
Empirical	Student ka exam pass karna	124 / 200	0.62
Theoretical	Student ID (1 se 200) random chuni, ID 5 se divisible (5, 10, ..., 200)	40 / 200	0.20

Farak: Empirical value data badalne par badal jaati hai (62%), jabki theoretical value fix rehti hai (20%) kyunki wo sirf counting par based hai.

Task 3: Random Variable & Probability Distribution

Random variable X = 3 random chune gaye students mein se kitne exam pass karenge. X ki values: 0, 1, 2 ya 3.

Har student ke pass hone ki probability p = 0.62 hai aur students independent hain, isliye X **Binomial(n = 3, p = 0.62)** follow karta hai:

$P(X = k) = C(3, k) \times p^k \times (1 - p)^{3-k}$ (q = 1 - p = 0.38)

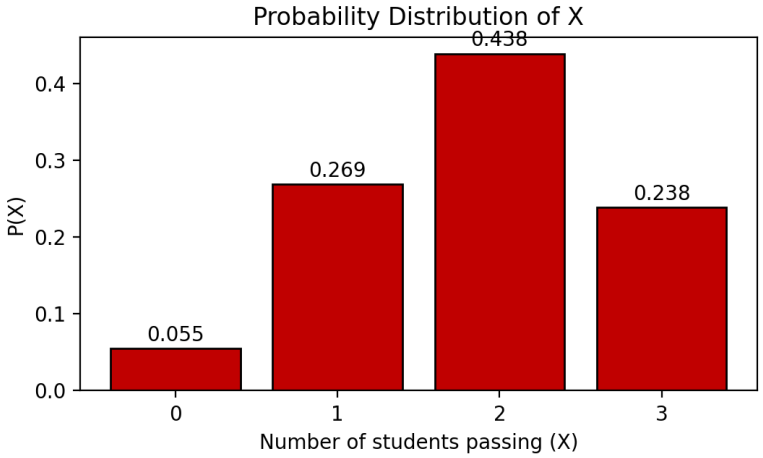
X	Calculation	P(X)
0	$C(3,0) \times 0.62^0 \times 0.38^3 = 1 \times 1 \times 0.0549$	0.0549
1	$C(3,1) \times 0.62^1 \times 0.38^2 = 3 \times 0.62 \times 0.1444$	0.2686
2	$C(3,2) \times 0.62^2 \times 0.38^1 = 3 \times 0.3844 \times 0.38$	0.4382
3	$C(3,3) \times 0.62^3 \times 0.38^0 = 1 \times 0.2383 \times 1$	0.2383
Total		**1.0000**

Mean aur Variance

Mean E[X] = sum of (X x P(X)) = 0(0.0549) + 1(0.2686) + 2(0.4382) + 3(0.2383) = **1.86**
Shortcut check: n x p = 3 x 0.62 = 1.86

Variance = $E[X^2] - (E[X])^2$, jahan $E[X^2]$ = 0(0.0549) + 1(0.2686) + 4(0.4382) + 9(0.2383) = 4.1664
Isliye Variance = 4.1664 - 3.4596 = **0.7068**
Shortcut check: n x p x q = 3 x 0.62 x 0.38 = 0.7068 (Std. deviation = 0.84)

Matlab: 3 students mein se average 1.86 (lagbhag 2) pass honge. Sabse zyada likely outcome **X = 2** hai (43.8%).

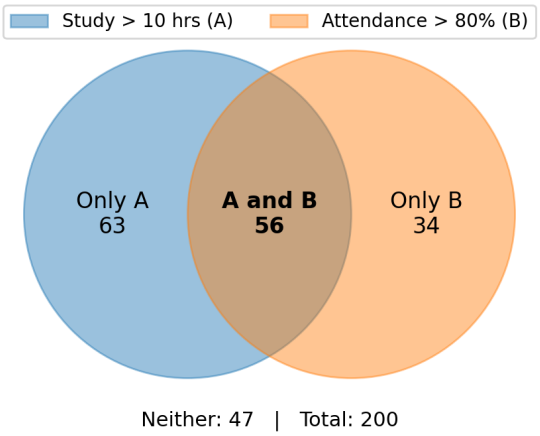


Task 4: Venn Diagram

Do sets lete hain: **A** = jo 10 ghante/week se zyada padhte hain (119 students), **B** = jinki attendance 80% se zyada hai (90 students). Dono ka overlap A and B wo students hain jo **dono conditions** poori karte hain.

Region	Count	Probability
Only A (sirf study > 10 hrs)	63	0.315
Only B (sirf attendance > 80%)	34	0.170
A and B (dono)	56	0.280
Neither (koi nahi)	47	0.235
Total	**200**	**1.000**

Check: |A| = 63 + 56 = 119 | |B| = 34 + 56 = 90 | |A or B| = 63 + 34 + 56 = 153. Union rule: P(A or B) = P(A) + P(B) - P(A and B) = 0.595 + 0.450 - 0.280 = 0.765.



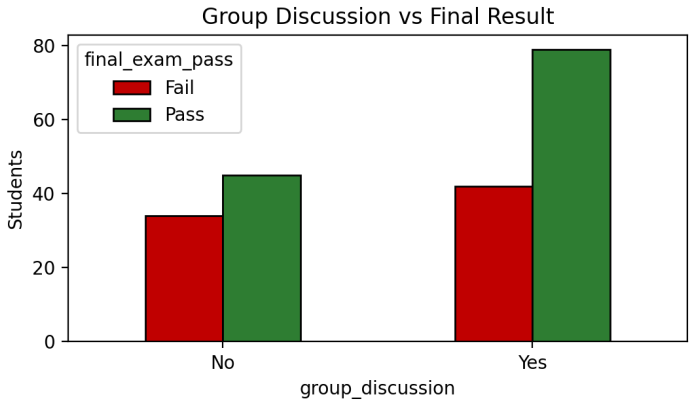
Task 5: Contingency Table & Probabilities

Contingency table do categorical variables (group discussion aur exam result) ko ek saath count karti hai:

Group Discussion	Pass	Fail	Total
Yes	79	42	121
No	45	34	79
Total	**124**	**76**	**200**

Type	Matlab	Formula and calculation	Result
Joint	Participate kiya AND pass hua	79 / 200	0.395
Marginal	Pass hua (kisi bhi group ka)	124 / 200	0.620
Conditional	Pass hua, *given* ki participate kiya tha	P(Pass and Yes) / P(Yes) = 79 / 121	0.6529

Conditional ko formula se bhi check kiya: P(Yes) = 121/200 = 0.605, to 0.395 / 0.605 = **0.6529**.



Task 6: Relationships samajhna

Conditional probability ka intuition

P(Pass | Yes) ka sawaal hai: "*Agar mujhe pehle se pata hai ki student group discussion mein participate karta hai, to uske pass hone ke kitne chances hain?*" Yahan hum 200 students ki jagah sirf **121 participating students** dekhte hain aur gin-te hain unme se kitne pass hue (79). Result 65.29% hai, jo overall pass rate (62%) se thoda zyada hai, yaani participation se thoda fayda dikhta hai.

Independent, Dependent ya Mutually Exclusive?

Test	Rule	Hamare numbers	Result
Mutually exclusive	P(A and B) = 0	P(Yes and Pass) = 0.395	Nahi (kyunki 0 nahi hai)
Independent	P(A and B) = P(A) x P(B)	0.605 x 0.62 = 0.375 vs 0.395	Lagbhag barabar
Independent	P(B | A) = P(B)	0.6529 vs 0.62	Lagbhag barabar
Chi-square test	p-value > 0.05 = independent	p-value = 0.2997	Independent

Conclusion: Group discussion aur exam pass karna **independent events** hain (is sample mein). Chhota sa farak (0.395 vs 0.375) sirf sampling noise hai, kyunki chi-square test ka p-value 0.30 hai jo 0.05 se bada hai. Dono events **mutually exclusive nahi** hain, kyunki ek student participate bhi kar sakta hai aur pass bhi ho sakta hai.

Task 7: Bayes' Theorem

Given: H = high attendance (> 80%). P(H | Pass) = 0.70, P(H | Fail) = 0.40, P(H) = 0.60.
Dhoondna hai: P(Pass | H).

Bayes' formula: P(Pass | H) = [P(H | Pass) x P(Pass)] / P(H)

Step 1: P(Pass) nikalna. Question mein P(Pass) diya nahi hai, isliye total probability se nikalte hain. Maan lo P(Pass) = x:

P(H) = P(H | Pass) x x + P(H | Fail) x (1 - x)
0.60 = 0.70x + 0.40(1 - x) = 0.40 + 0.30x
x = (0.60 - 0.40) / 0.30 = **2/3 = 0.6667**

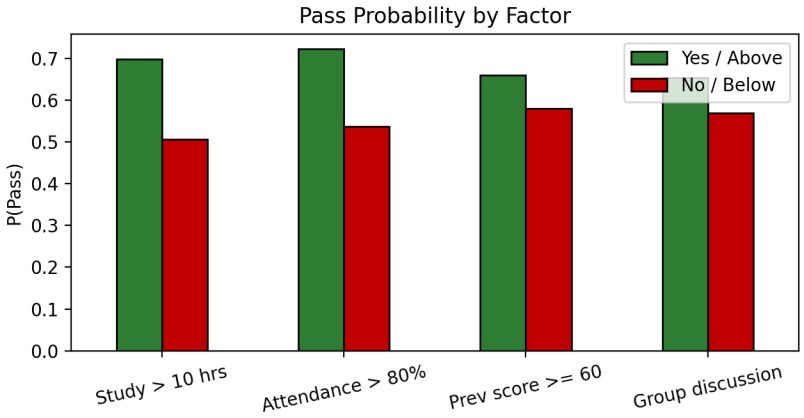
Step 2: Bayes lagana.

P(Pass | H) = (0.70 x 0.6667) / 0.60 = 0.4667 / 0.60 = **0.7778 (77.78%)**

Matlab: Jis student ki attendance high hai, uske pass hone ke chances lagbhag **77.8%** hain, jabki bina kuch jaane pass hone ke chances 66.7% the. Yaani high attendance se probability badhti hai.

Kaun sa factor sabse zyada asar karta hai?

Factor	Pass rate (Yes/Above)	Pass rate (No/Below)	Difference	Correlation
Study > 10 hrs/week	69.7%	50.6%	**+19.1%**	0.230
Attendance > 80%	72.2%	53.6%	**+18.6%**	0.206
Previous score >= 60	66.0%	58.0%	+8.0%	0.209
Group discussion	65.3%	57.0%	+8.3%	(independent)



Final Summary

- 200 students mein se **124 pass (62%)** aur 76 fail (38%) hue.
- **Study hours** sabse bada factor hai: 10 ghante/week se zyada padhne wale 69.7% pass hue, baaki sirf 50.6%.
- **Attendance** lagbhag utni hi important hai: 80% se upar wale 72.2% pass hue vs 53.6%. Bayes ke hisaab se bhi P(Pass | high attendance) = 77.8%.
- **Previous test score** ka positive lekin chhota asar hai; sirf purane marks se result decide nahi hota.
- **Group discussion** ka asli asar bahut kam hai; chi-square test (p = 0.30) ke hisaab se wo pass hone se independent hai.
- 3 random students mein expected passes **1.86**, variance **0.707**; sabse likely outcome exactly 2 pass (43.8%).
- Lagbhag **28% students (56)** dono karte hain: 10+ ghante padhna aur 80%+ attendance.

Conclusion: Pass hone ki probability sabse zyada **study hours** aur **attendance** se tay hoti hai, uske baad previous test score. Group discussion ka khaas asar nahi dikha. Institute ko students ko regular padhai (10+ hrs/week) aur high attendance (80%+) ke liye encourage karna chahiye. *Note: Yeh AI-generated sample data hai, isliye conclusions sirf isi dataset par lagu hote hain.*